Elyas Demamu

Software Developer | AI Systems Builder | Technology Innovator

Location: Addis Ababa, Ethiopia

Father Name: Demamu

PROFESSIONAL SUMMARY

I am an independent software developer and technology enthusiast passionate about building intelligent software systems, AI-powered applications, and practical digital solutions.

My work focuses on combining software engineering, artificial intelligence, automation, and modern web technologies to create useful, scalable, and user-centered products.

I have hands-on experience designing and developing full-stack applications, building backend APIs, integrating AI providers, working with databases, implementing authentication and security systems, and deploying applications to production environments.

I am particularly interested in AI agents, autonomous systems, intelligent assistants, AI infrastructure, and the future of human-computer interaction.

I approach development with a systems-thinking mindset: understanding the problem, designing the architecture, implementing the solution, testing its reliability, and continuously improving the product.

CORE TECHNICAL SKILLS

Programming & Development

- TypeScript and JavaScript
- React and Next.js
- Node.js and Express.js
- Full-stack application development
- REST API development and integration

Backend & Database Engineering

- PostgreSQL
- Prisma ORM
- Database architecture and migrations
- Authentication and authorization
- API architecture and security

Artificial Intelligence

- AI API and provider integration
- Multi-provider AI architectures
- AI assistants and conversational systems
- AI context and memory systems
- AI agent workflows and automation
- AI-powered study and productivity tools
- Model selection and provider routing

Tools & Infrastructure

- Git and GitHub
- Docker
- Render
- Neon PostgreSQL
- OpenAPI
- Visual Studio Code

- Playwright and automated testing
- Windows and PowerShell development environments

FEATURED PROJECTS

1. Tutor Walle — AI-Powered Academic Platform

Role: *Independent Developer | Full-Stack & AI Systems Development*

Designed and developed an academic productivity platform that combines student management, learning resources, academic planning, and artificial intelligence.

Key contributions:

- Developed a full-stack application using Next.js, TypeScript, Express, and PostgreSQL.
- Designed academic management systems for courses, semesters, tasks, examinations, notes, and learning resources.
- Integrated AI assistants with contextual understanding of academic information.
- Implemented AI-powered study planning and generation workflows.
- Developed AI provider integrations and model-routing capabilities.
- Implemented authentication, database persistence, file uploads, and security controls.
- Built and tested production deployment infrastructure using Render and Neon.

Technologies: Next.js, React, TypeScript, Express.js, Prisma, PostgreSQL, Docker, Render, GitHub.

2. AI Command Center — Intelligent AI Management Platform

Role: *Project Architect & Developer*

Working on an AI management platform designed to bring AI capabilities, tools, and workflows into one unified interface.

Key focus areas:

- Modern React and TypeScript application architecture.
- AI system management and integration.
- Interactive user interfaces and responsive dashboards.
- Automated browser testing and quality assurance.
- Designing a unified environment for managing AI capabilities.

Technologies: React, TypeScript, Vite, Playwright.

3. Solopreneur's Influence Academy — Intelligent Learning Platform

Role: *Project Concept & System Architect*

Developing the system concept and technical direction for an integrated educational platform focused on understanding human influence, psychology, persuasion, negotiation, and critical thinking.

Key focus areas:

- Designing a structured academic learning experience.
- Integrating research-based educational content.
- Developing learning pathways and curriculum architecture.
- Exploring intelligent study assistance and personalized learning.
- Creating an integrated platform rather than a conventional content website.

4. Personal AI Capability Network — AI Agent Ecosystem

Role: *Independent Research & System Design*

Exploring an interconnected AI ecosystem designed to coordinate different AI models, agents, tools, and specialized capabilities.

Areas of interest:

- Multi-agent architectures.
- AI orchestration and delegation.
- Specialized AI agents.
- Automated workflows.

- Intelligent tool integration.
- Unified AI capability management.

PROFESSIONAL INTERESTS

- Artificial Intelligence and Machine Learning
- Autonomous AI Agents and Multi-Agent Systems
- Full-Stack Software Engineering
- AI Infrastructure and Orchestration
- Automation and Intelligent Workflows
- Human-Computer Interaction
- Educational Technology
- Emerging Technologies and Innovation

DEVELOPMENT APPROACH

I believe in building technology that solves real problems.

My development philosophy emphasizes:

- Understanding systems before implementing them.
- Building maintainable and scalable architectures.
- Combining practical functionality with good user experience.
- Prioritizing security and reliability.
- Testing software beyond simply making it work.
- Continuously learning and experimenting with emerging technologies.

CAREER OBJECTIVE

My goal is to contribute to the development of intelligent software products and next-generation AI systems.

I am particularly interested in opportunities involving software engineering, artificial intelligence, AI agents, automation, and innovative technology development.

I aim to build technology that makes complex tasks simpler, expands human capabilities, and creates meaningful real-world impact.

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